

TASK- 4 B

(Both Esp1 and Esp2 acts as receiver and transmitter)

One Esp is connected to ultrasonic sensor (Esp1) and other esp is connected to a relay module(Esp2).The distance measured by ultrasonic sensor is send to the the Esp2 using EspNow protocol when the distance is within the threshold the relay is turned on the command of the current status of relay is also send back to the ultrasonic sensor so here both esp1 and esp2 acts receiver and transmitter

NOTE : Esp1 and Esp2 are connected to different laptops also we can do it in a single laptop

Procedure

1. Create a new file and type Ctrl + shift + s
2. Select micropython device option
3. Save as [boot.py](#)
4. Create another file and type ctrl+shift+s and save this also in micropython device and save it as [main.py](#)
5. Both esp1and esp2 should be connected to the usb ports (Com3 and com4)
6. In [boot.py](#) write the code for Esp1 and select com3 port save and run it
7. Without stopping it go to [main.py](#) and change the port from com3 to com4 and then write the code for Esp2 and save it and run

“ But the problem with using a single laptop is we can't see both the receiving and transmitting command of both esp's simultaneously ”

It is better to use two laptops to see how it actually works

CODE FOR GETTING MAC ADDRESS OF BOTH ESP

```
import network

# Activate wireless interface

w0 = network.WLAN(network.STA_IF)

w0.active(True)

# Print MAC Address in standard hex format

mac = w0.config('mac')

print("Receiver MAC Address:", ':'.join(['%02x' % b for b in mac]))
```

Note down the printed address (e.g., 24:0a:c4:xx:xx:xx). You will need to paste this string into the transmitter script.

NOTE :

Mac address of esp1 is given to the code of esp2 and that of esp2 is given to code for esp1

ESP32 #1: Ultrasonic Sensor

Wiring: Trig -Pin 5 &Echo - Pin 18

```
import machine
import time
import network
import espnow # Using the standard module directly

# 1. Setup Pins for HC-SR04
trigger = machine.Pin(5, machine.Pin.OUT)
echo = machine.Pin(18, machine.Pin.IN)

# 2. Setup Wi-Fi and ESP-NOW
wlan = network.WLAN(network.STA_IF)
wlan.active(True)
wlan.disconnect() # Essential for ESP-NOW to run smoothly

e = espnow.ESPNow()
e.active(True)

# !!! REPLACE WITH YOUR ESP32 #2 (RELAY BOARD) MAC ADDRESS !!!
RELA_BOARD_MAC = b'\x24\x0a\xc4\x00\x11\x22'
e.add_peer(RELA_BOARD_MAC)

def get_distance():
    trigger.value(0)
    time.sleep_us(2)
    trigger.value(1)
    time.sleep_us(10)
    trigger.value(0)

    # Simple timeout loop to prevent infinite freezing if sensor
    disconnects
    timeout = time.ticks_us() + 30000
    while echo.value() == 0:
        signaloff = time.ticks_us()
        if signaloff > timeout: return 0
```

```

while echo.value() == 1:
    signalon = time.ticks_us()

    timepassed = signalon - signaloff
    distance = (timepassed * 0.0343) / 2
    return distance

print("Sensor Board Initialized. Starting loop...")

while True:
    dist = get_distance()
    print("\n--- New Reading ---")
    print("Measured Distance: {:.2f} cm".format(dist))

    # 2. Send data string to ESP32 #2
    msg = str(dist)
    try:
        e.send(RELA_BOARD_MAC, msg)
        print(" -> VERIFICATION: Distance data packet pushed to Relay
ESP32.")
    except Exception as err:
        print(" -> ERROR: Send failed:", err)

    # 3. Check for the return acknowledgement payload from the relay
board
    # Timeout set to 150 milliseconds
    host, recv_msg = e.recv(150)
    if recv_msg:
        status_report = recv_msg.decode('utf-8')
        print(" <- ACKNOWLEDGEMENT: Received statement back from Relay
Board!")
        print("      Status Report:", status_report)

    time.sleep(1)

```

ESP32 #2: Relay Control

Wiring: Relay - Pin 25

```

import machine
import network
import espnow
import time

# 1. Setup Relay Pin
relay = machine.Pin(25, machine.Pin.OUT)

```

```

relay.value(0)

# 2. Setup Wi-Fi and ESP-NOW
wlan = network.WLAN(network.STA_IF)
wlan.active(True)
wlan.disconnect()

e = espnow.ESPNow()
e.active(True)

# !!! REPLACE WITH YOUR ESP32 #1 (SENSOR BOARD) MAC ADDRESS !!!
SENSOR_BOARD_MAC = b'\x24\x0a\xc4\xaa\xbb\xcc'
e.add_peer(SENSOR_BOARD_MAC)

print("Relay Board Initialized. Waiting for data...")

while True:
    # Read incoming data (This will wait here until data arrives)
    host, msg = e.recv()

    if msg:
        try:
            distance_received = float(msg.decode('utf-8'))
            print("\n<- CHECKPOINT: Signal received from Sensor
ESP32!")
            print("    Extracted Distance: {:.2f}
cm".format(distance_received))

            # Check range threshold (20cm to 30cm)
            if distance_received <= 30.0:
                relay.value(1) # Turn Relay ON
                print("    [!] Target Range Met! Physical Relay
Switched ON.")
                reply = "RELAY IS ON: Target within 20-30cm range."
            else:
                relay.value(0) # Turn Relay OFF
                print("    [ ] Out of Range. Physical Relay Switched
OFF.")
                reply = "RELAY IS OFF: Outside target range."

            # Send the verification statement back to ESP32 #1
            e.send(SENSOR_BOARD_MAC, reply)
            print(" -> TELEMETRY: Status response sent back.")

        except ValueError:
            print(" -> ERROR: Received a corrupted data payload.")

```

OUTPUT

THONNY

```
Thonny - C:\Users\ANAGHA\main.py @ 12:40
File Edit View Run Tools Help
main.py
7 TRIGGER_PIN = 5
8 ECHO_PIN = 18
9 THRESHOLD_CM = 30
10
11 # REPLACE THIS with your Relay ESP's MAC address bytes format
12 RELAY_MAC = b'\x00\x4b\x12\xef\x06\x20'
13
14 # Setup Ultrasonic Pins

Shell
Distance: 44.8 cm
Distance: 44.9 cm
Distance: 44.9 cm
Distance: 44.9 cm
Distance: 44.8 cm
Distance: 44.9 cm
Distance: 43.4 cm
Distance: 43.0 cm
Distance: 43.4 cm
Distance: 43.4 cm
Distance: 43.4 cm
Distance: 43.5 cm
Distance: 43.4 cm
Distance: 17.1 cm
Object detected! Sending ON command...
Feedback Received from Relay: Relay is ON
Distance: 273.7 cm
Distance: 999.0 cm
Distance: 5.3 cm
Object detected! Sending ON command...
Feedback Received from Relay: Relay is ON

MicroPython (ESP32) • COM3
30°C Partly sunny 12:26 PM 02-06-2026
```

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